PUB. DIM-1358-000

# Live Link Plugin for Unreal Engine

(Windows / macOS / Linux)

ご使用の前に、必ず本書をよくお読みのうえ、正しくお使いください。
また、本書はいつでもご覧になれるよう、大切に保管してください。

Before using the product, carefully read this document and be sure to use the product correctly.  Retain this document for future reference.

**使用説明書**

**Instructions**

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# Introduction

This plugin allows Unreal Engine (game engine) to handle CV Metadata files output in real time from a Canon camera (hereafter referred to as "camera") equipped with a Canon lens (hereafter referred to as "lens") that supports virtual production. For information regarding system requirements and supported lenses and cameras, refer to your local Canon website.

# About this Manual

Symbols and icons used in this document:

| | |
|---|---|
| 🕮 | Reference page number in this document. |

- Descriptions use Windows 10 for sample screens. Screenshots used are from a product in development and may differ slightly from the actual screens due to product enhancement.
- Unreal Engine descriptions in this document use Unreal Engine 5.3.2 menu names and screenshots.
- Some screenshots have been altered to make them easier to read.

# Camera/Unreal Engine Setup Preparations

In this document it is assumed that this software has been downloaded, and that Unreal Engine and an Integrated Development Environment (IDE) recommended for Unreal Engine is installed correctly on your computer. Attach a lens that supports virtual production to the camera (for details, refer to the manual of the camera). Use the 品 (Ethernet) terminal on the camera to connect to the same network as the computer on which Unreal Engine is installed.

## Camera Setup

Refer to "CV Protocol" on the camera's manual when performing this procedure.

1. Use MENU > [Network Settings] > [New Conn. Setting (Wizard)] and save the [CV Protocol] communication and function settings to the desired SET option.
2. Select MENU > [Network Settings] > [Connect] > the SET where [CV Protocol] was saved.
3. Select [OK].
   - When connection to the network is completed, CV Metadata is output from the 品 (Ethernet) terminal.

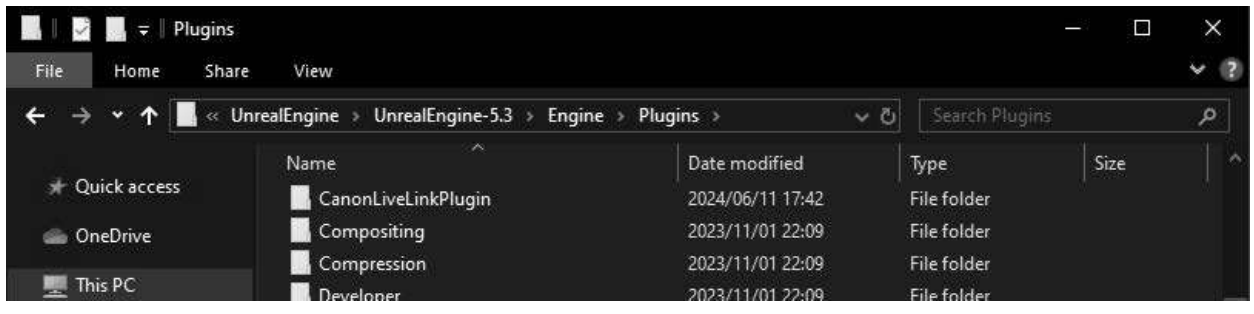
## Unreal Engine Setup

For details on how to operate Unreal Engine, refer to the Unreal Engine instruction manual.

### ■ Enabling the plugin

1. Start Unreal Engine.

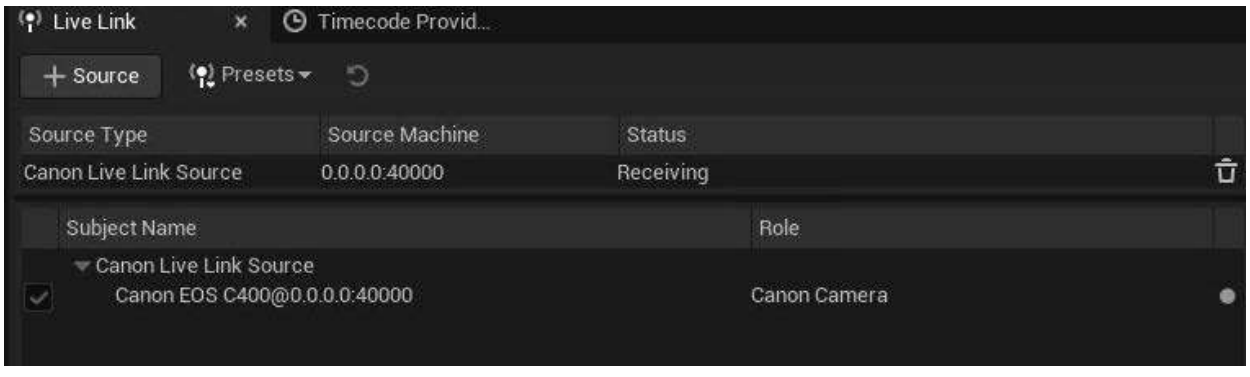
2. Create a new project, then copy the downloaded "CanonLiveLinkPlugin" folder to the "Plugins" folder of the created project.




- In case there is no “Plugins” folder in the project, create it manually.
- When using the plugin with other projects, copy the downloaded “CanonLiveLinkPlugin” folder to the “Plugins” folder of each project.
- You can also copy the "CanonLiveLinkPlugin" folder to the Unreal Engine folder and operate the Integrated Development Environment (IDE) to build the plugin.

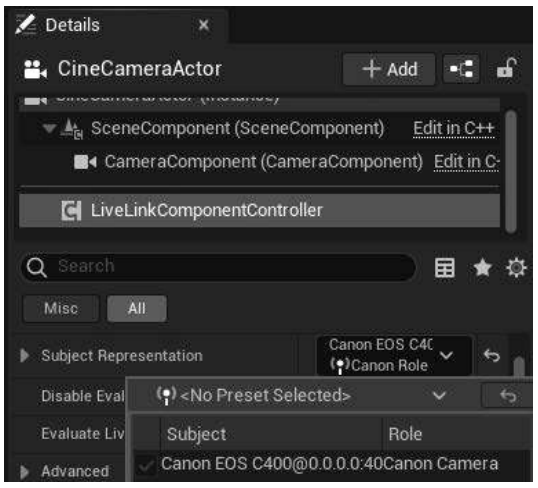
3. Restart Unreal Engine and build the plugin.
   - After restarting, follow the onscreen instructions to build the plugin.
4. Select [Edit] > [Plugin] on the menu to open the plugin window.
5. Make sure that each [CanonLiveLinkPlugin], [Live Link], [LiveLinkCamera], [LiveLinkLens] and [Camera Calibration] plugin is enabled.

## ■ Adding a subject (supported camera)



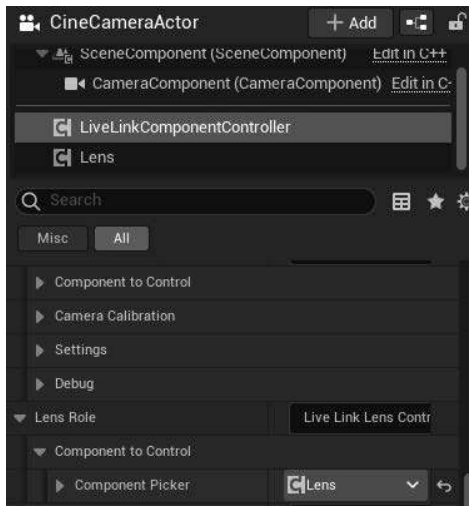

1. Open the [Live Link] window and add a source.
   - Click the [+Source] button and select [Canon Live Link Plugin].
2. Enter the IP address and port number of the camera you want to connect to.
   - Enter the IP address of the [Destination Server] set for [CV Protocol] on the camera’s [ Network Settings] menu.
   - For the port number, enter the [Dest. Port No.] set for [CV Protocol] on the camera's [ Network Settings] menu.
3. Click [OK].
   - When camera connections are complete, the [Subject Name] field will display the name of the camera connected as a subject. Hereafter, the camera in the [Subject Name] column will be described as “subject”.

## ■ Adding a Cine Camera Actor



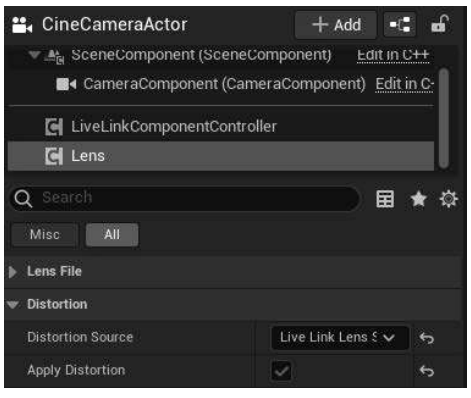

1. Select [Cinematic] > [CineCameraActor] from  in the toolbar, then add a Cine Camera Actor.
   - The [CineCameraActor] will appear in the [Outliner] window. If a Cine Camera Actor has been added before, a number is added at the end.
2. Select a Cine Camera Actor in the [Outliner] window.
   - The selected Cine Camera Actor will appear in the [Details] tab.
3. Add a component to the Cine Camera Actor.
   - Click the [+Add] button in the [Details] tab and select [Live Link Controller].
4. Set [LiveLinkComponentController] in the subject.
   - Select the added subject from [Subject Representation] in the [Details] tab (📖 4).

## ■ Applying distortion to a Cine Camera Actor




1. Add a component to the Cine Camera Actor.
   - Click the [+Add] button in the [Details] tab and select [Lens].

2. Select the [LiveLinkComponentController] component.
3. Select [Lens Role] > [Component to Control] > [Component Picker] > [Lens] in the [Details] tab.
4. Select the [Lens] component.




5. Select [Distortion Source] > [Live Link Lens Subject] in the [Details] tab, then enable the [Apply Distortion] checkbox.

## ■ Setting [Update Flags]

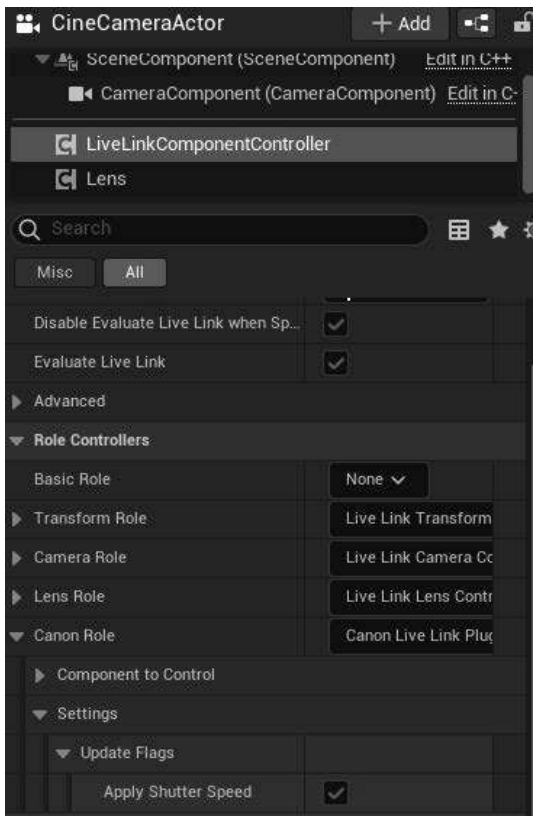
Set the CV Metadata items to be output from the camera in the subject. All checkboxes are enabled by default.

**Items other than Shutter Speed:**

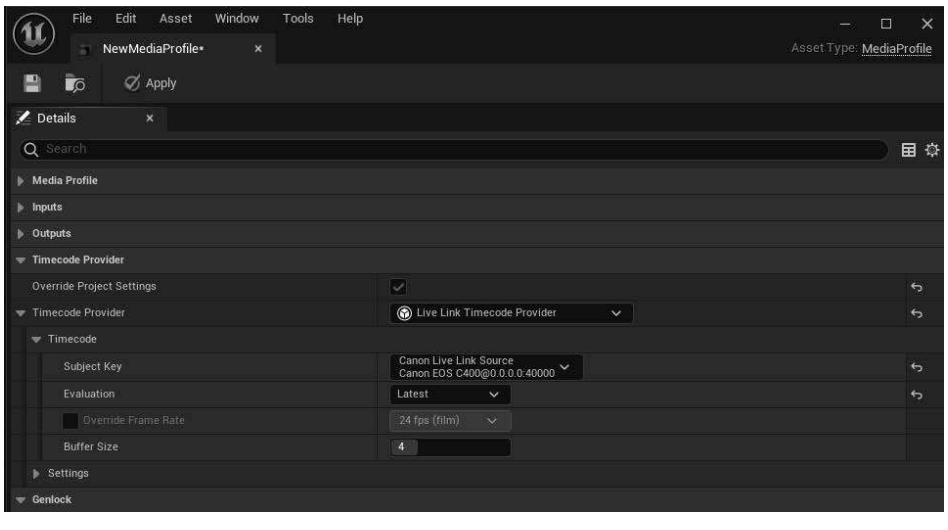
Select [Camera Role] > [Settings] > [Update Flags] in the [Details] tab, and uncheck the items you do not wish to link to the subject.

**Shutter Speed:**

Select [Canon Role] > [Settings] > [Update Flags] in the [Details] tab, and uncheck this item in case you do not wish to link it to the subject.



■ Applying the time code

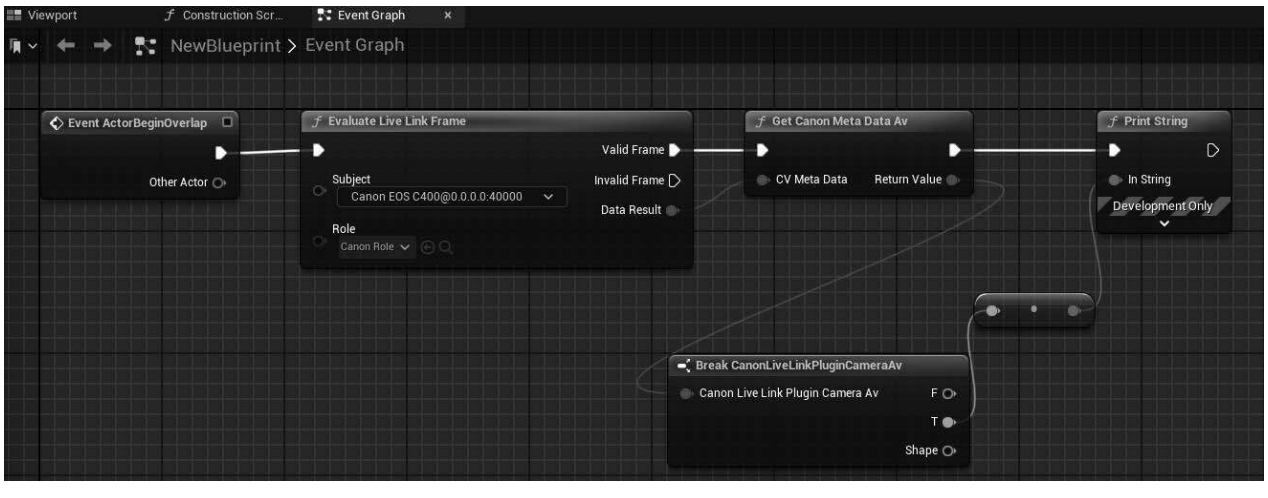



1. Right-click [Content Browser] at the bottom left of the screen and select [Media] > [Media Profile].
2. Enable the [Timecode Provider] > [Override Project Settings] checkbox in the [Details] tab.
3. Select [Timecode Provider] > [Live Link Timecode Provider].
4. Select the added subject (📖 4) from [Timecode Provider] > [Time Code] > [Subject Key].
5. Select [Evaluation] > [Nearest] or [Latest].

■ Retrieving metadata using Blueprint functions

CV Metadata output from the camera can be retrieved using the Blueprint function.

Reference example:



# Others

## ■ Metadata list

The following table lists the parameters that are output from the camera, and which parameters within the list are applied to the Cine Camera Actor. Non-applied parameters can be accessed using Blueprint functions.

| Ethernet Metadata | | | Applied parameters | Setting switching method |
|---|---|---|---|---|
| Key 1 | Key 2 | Key 3 | | |
| version | - | - | - | - |
| camera | model | - | Subject Name | - |
| | serial | - | - | - |
| | ver | - | - | - |
| lens | name | - | - | - |
| | serial | - | - | - |
| | ver | - | - | - |
| sensor | size | - | - | - |
| | pix | - | - | - |
| | fps | - | - | - |
| | expTime | - | - | - |
| picture | pix | - | - | - |
| | fps | - | Timecode<br>Shutter Speed | Media Profile<br>Canon Role |
| | scan | - | - | - |
| | timecode | - | Timecode | Media Profile |
| | subFrame | - | Timecode | Media Profile |
| | dropframe | - | Timecode | Media Profile |
| cameraIdx | - | - | - | - |
| av | f | - | Current Aperture | Camera Role |
| | t | - | - | - |
| | shape | - | - | - |
| tv | angle | - | Shutter Speed | Canon Role |
| sv | iso | - | - | - |
| nd | - | - | - | - |
| focalLength | - | - | - | - |
| focusDistance | - | - | Current Focus Distance | Camera Role |
| matchMvData | - | - | - | - |
| tilt | - | - | - | - |
| roll | - | - | - | - |
| colorTemp | - | - | - | - |
| tint | - | - | - | - |
| correction | distortion | - | - | - |
| | shading | - | - | - |
| | chromaAberration | - | - | - |
| | focusBreathing | - | - | - |
| entPupilPos | - | - | Current Focus Distance | Camera Role |

| | | | | |
|---|---|---|---|---|
| cameraCarib | sensorSize | - | Filmback<br>Current Horizontal FOV | Camera Role |
| | flOutImg | - | Current Focal Length<br>Current Horizontal FOV | Camera Role |
| | opencv | flPix | Distortion State | Apply Distortion |
| | | centerPix | Distortion State | Apply Distortion |
| | | distortion | Distortion State | Apply Distortion |

## ■ Troubleshooting

| Problem | Cause/Solution | 🕮 |
|---|---|---|
| A compile error appears. | • There is no compilation environment configured on the computer. Build a compilation environment, restart the computer, then start Unreal Engine again. | _ |

# Trademarks


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